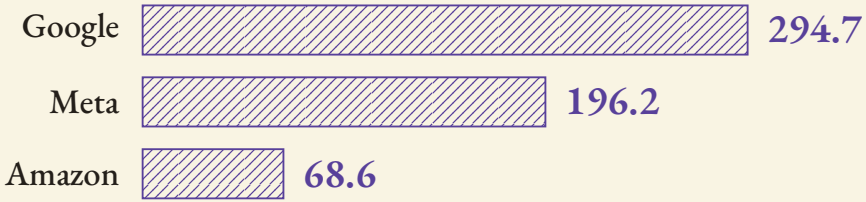
Three companies, 1.5 billion dollars of ads a day

Advertising revenue in 2025, in billions of US dollars

Alphabet and Meta: annual reports (Form 10-K) for 2025. Amazon: its results for the four quarters of 2025, added up.

Three companies, 1.5 billion dollars of ads a day

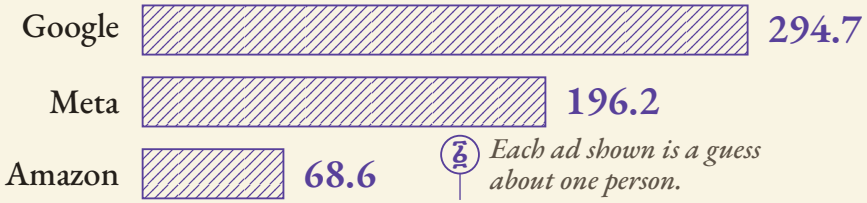
Advertising revenue in 2025, in billions of US dollars



Alphabet and Meta: annual reports (Form 10-K) for 2025. Amazon: its results for the four quarters of 2025, added up.

Three companies, 1.5 billion dollars of ads a day

Advertising revenue in 2025, in billions of US dollars



Together **559.5** billion, about **1.5** billion a day

Alphabet and Meta: annual reports (Form 10-K) for 2025. Amazon: its results for the four quarters of 2025, added up.

A machine suggested much of what we watch and buy

What share of your hours on YouTube did you go looking for?

A machine suggested much of what we watch and buy

What share of your hours on YouTube did you go looking for?

YouTube, watch time

Neal Mohan at CES 2018, in CNET

Share that came from a recommendation



over 70%

A machine suggested much of what we watch and buy

What share of your hours on YouTube did you go looking for?

Share that came from a recommendation

YouTube, watch time

Neal Mohan at CES 2018, in CNET



over 70%

Netflix, viewing

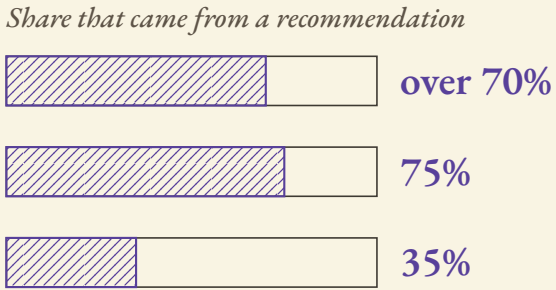
Netflix Tech Blog, 2012



75%

A machine suggested much of what we watch and buy

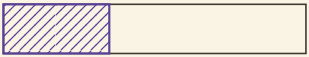
What share of your hours on YouTube did you go looking for?



The question to ask: compared to what?

Through a recommendation

Share of Amazon purchases that came from a recommendation



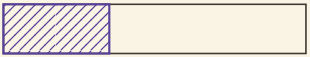
35%

McKinsey, 2013.

The question to ask: compared to what?

Through a recommendation

Share of Amazon purchases that came from a recommendation

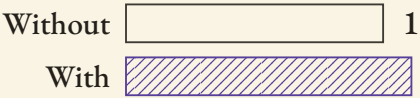


35%

McKinsey, 2013.

Caused by a recommendation

Randomized experiment: sales of a product and its recommended substitutes, on average



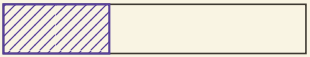
+11% more sales

Kumar and Hosanagar, Information Systems Research, 2019. Jannach and Jugovac, ACM TMIS, 2019.

The question to ask: compared to what?

Through a recommendation

Share of Amazon purchases that came from a recommendation



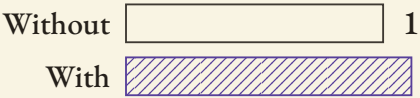
35%

A share also counts purchases that would have happened anyway.

McKinsey, 2013.

Caused by a recommendation

Randomized experiment: sales of a product and its recommended substitutes, on average



+11% more sales

Published field tests more often report 1% to 5% more revenue.

Kumar and Hosanagar, Information Systems Research, 2019. Jannach and Jugovac, ACM TMIS, 2019.

Six customers, two calls: whom do you call?

Invented customers

Customer	Chance of leaving	If we call	Change
Lea	80%		
Marco	70%		
Nina	50%		
Jonas	40%		
Sara	30%		
Luca	10%		

Six customers, two calls: whom do you call?

Invented customers

Customer	Chance of leaving	If we call	Change
Lea	80%		
Marco	70%		
Nina	50%		
Jonas	40%		
Sara	30%		
Luca	10%		

Called by risk

Six customers, two calls: whom do you call?

Invented customers

Customer	Chance of leaving	If we call	Change
Lea	80%	80%	0
Marco	70%	75%	+5
Nina	50%	30%	−20
Jonas	40%	25%	−15
Sara	30%	25%	−5
Luca	10%	10%	0

Called by risk

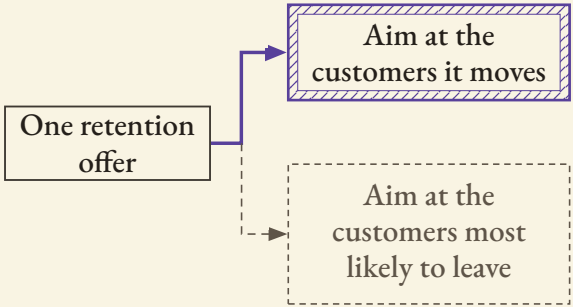
Six customers, two calls: whom do you call?

Invented customers

Customer	Chance of leaving	If we call	Change	
Lea	80%	80%	0	Called by risk
Marco	70%	75%	+5	
Nina	50%	30%	−20	Called by response
Jonas	40%	25%	−15	
Sara	30%	25%	−5	
Luca	10%	10%	0	

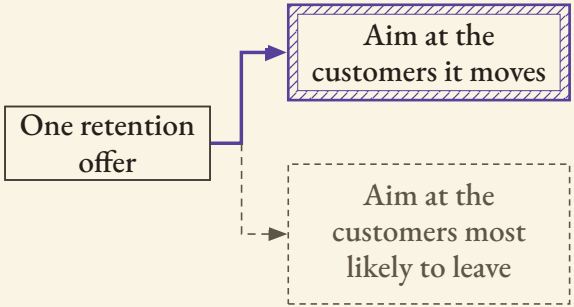
By risk, nothing changes for Lea and Marco becomes more likely to cancel.

Call the customers the offer can move



Eva Ascarza, Journal of Marketing Research, 2018: two field experiments, at a membership organization in North America and a telecom firm in the Middle East. The numbers as summarized by the American Marketing Association, 2023; the comparison was simulated on the field data.

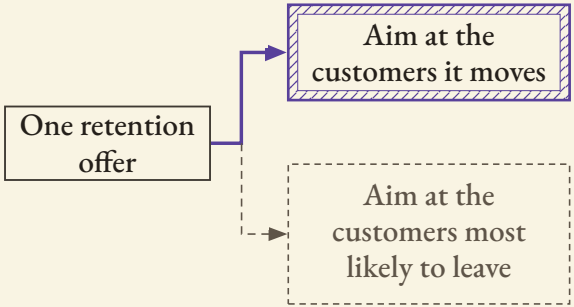
Call the customers the offer can move



4.1 to 8.7 percentage points less churn than aiming by risk

Eva Ascarza, Journal of Marketing Research, 2018: two field experiments, at a membership organization in North America and a telecom firm in the Middle East. The numbers as summarized by the American Marketing Association, 2023; the comparison was simulated on the field data.

Call the customers the offer can move



4.1 to 8.7 percentage points less churn than aiming by risk

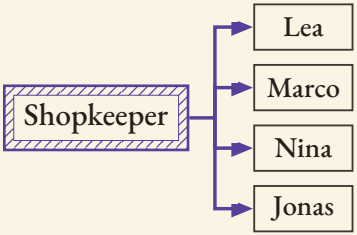
In some cases, aiming at them raised churn

Eva Ascarza, Journal of Marketing Research, 2018: two field experiments, at a membership organization in North America and a telecom firm in the Middle East. The numbers as summarized by the American Marketing Association, 2023; the comparison was simulated on the field data.

What the shopkeeper knew, a machine must learn

The village shop

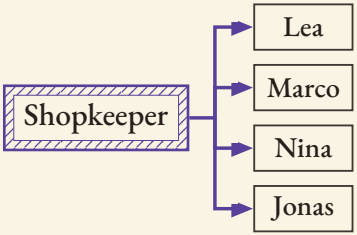
She knows every regular



What the shopkeeper knew, a machine must learn

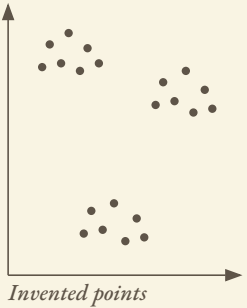
The village shop

She knows every regular



Millions of regulars

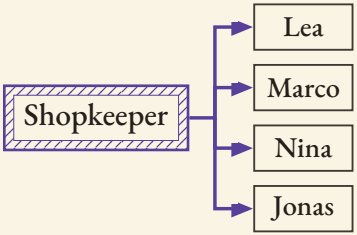
Loyalty cards such as Migros Cumulus and Coop Supercard



What the shopkeeper knew, a machine must learn

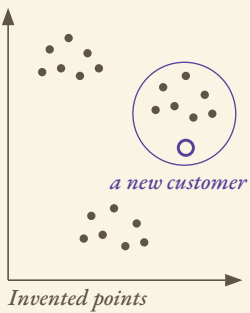
The village shop

She knows every regular



Millions of regulars

Loyalty cards such as Migros Cumulus and Coop Supercard

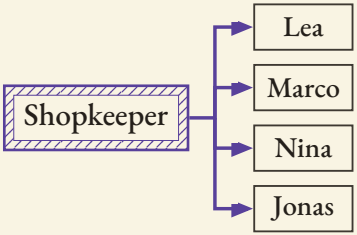


Customers who buy alike sit close together.

What the shopkeeper knew, a machine must learn

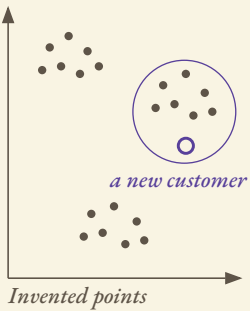
The village shop

She knows every regular



Millions of regulars

Loyalty cards such as Migros Cumulus and Coop Supercard



Customers who buy alike sit close together.

This weekend: how a machine learns where each customer sits.